PROJECT PROPOSAL | FINAL-YEAR ENGINEERING SPONSORED PROJECT

Smart Attendance System

Face & QR based, location-verified attendance for all Sheet Shapers plants

Project Understanding & Proposed Solution | Draft v1.1 for discussion

Prepared for	Sheet Shapers (sheetshapers.com)
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Status	Updated after our phone discussion (payable salary estimate added). Please mark changes, additions or removals and return to us.

How to use this document. Section 3 lists every requirement as we understood it, each with a reference number (R1, R2...). Please tick **OK**, **Change** or **Remove** against each one, and add anything we missed in Section 10. Section 8 lists questions that **must be answered by Sheet Shapers** before we start building.

1. Purpose of this document

Thank you for sponsoring our final-year project. During our visit you explained your attendance problems and what you would like. This document puts that conversation in writing so that both sides agree on **what we will build** before we start. It explains, in simple terms, what we understood, the solution we propose, and what we need from you.

2. What we understood

2.1 Your situation

- Sheet Shapers runs **several plants at different locations**.
- There are two kinds of workers: **permanent employees** and **temporary workers** (labour paid per hour, who may not come back next week or even tomorrow).
- Employees currently mark attendance through an app or website.

2.2 The main problem: proxy attendance

Some employees mark attendance even when they are **not present at the plant**, for example from home, or a colleague marks it for them. This leads to wrong attendance, wrong wages, and an unreliable count of who is actually inside the plant.

2.3 Why one method does not suit everyone

Face recognition works well for permanent employees, because their face is registered once and used for years. For temporary workers who change often, registering faces every time is slow and impractical. So temporary workers need a quicker method: a **personal QR code**.

3. Your requirements (please review)

Each line below is one requirement as we understood it. Please mark the last column.

Ref	Requirement	What it means in practice	OK / Change / Remove
R1	QR code attendance	Every temporary worker gets a unique QR code (for example on an ID card). Scanning it marks that worker's attendance.	[ ] OK [ ] Change [ ] Remove
R2	Face attendance	Permanent employees mark attendance by looking at a camera. The system recognises their face; no card or password needed.	[ ] OK [ ] Change [ ] Remove
R3	Stop proxy attendance	Attendance is accepted only when the person is truly present inside the plant. Marking from home or for someone else must not work.	[ ] OK [ ] Change [ ] Remove
R4	Geo-fencing	A virtual boundary is drawn around each plant on a map. Attendance from outside that boundary is rejected.	[ ] OK [ ] Change [ ] Remove
R5	Multiple plants	One system for all plants. Each plant has its own boundary, devices, shifts and reports; head office can see all plants together.	[ ] OK [ ] Change [ ] Remove
R6	Two employee types	Separate handling for permanent (face) and temporary/hourly workers (QR), including hours worked for hourly workers.	[ ] OK [ ] Change [ ] Remove

Ref	Requirement	What it means in practice	OK / Change / Remove
R7	Works with your hardware	Runs on devices you provide (tablet, phone, PC with camera). No special or costly attendance machine required.	[ ] OK [ ] Change [ ] Remove
R8	Website / app	Delivered as a website that can also be installed like an app on phones and tablets.	[ ] OK [ ] Change [ ] Remove
R9	No licence or usage fees	Built only with free, open-source technology. No paid APIs or subscriptions for you to pay later.	[ ] OK [ ] Change [ ] Remove
R10	Government compliance	Follows Indian rules on personal/biometric data and on attendance and working-hour records.	[ ] OK [ ] Change [ ] Remove
R11	Payable salary estimate	Using each employee's fixed salary structure and his attendance, the admin can see how much salary is payable for the month (see Section 5.1). Admin can enter deductions such as PF and TDS manually.	[ ] OK [ ] Change [ ] Remove
R12	Full payroll (future)	Automatic PF/ESIC/TDS calculation, statutory reports, challans, Form 16 and tax filing are **not** part of this version (see Section 5.2).	[ ] OK [ ] Change [ ] Remove

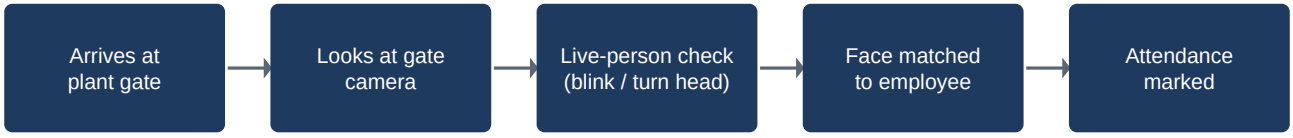
4. Our proposed solution

We propose a **Smart Attendance System**: one website with three parts, used by different people.

Part	Who uses it	What it does
Gate Attendance Screen (kiosk)	All workers, at the plant gate	A tablet or phone fixed at each plant gate, with its camera on. Permanent employees look at it; temporary workers show their QR card. Attendance is marked in about 2-3 seconds.
Employee App	Permanent employees (optional)	An app on the employee's own phone. Works only inside the plant boundary, only on the employee's own registered phone, and only after a live face check.
Admin Dashboard	HR, plant admins, supervisors, management	Add employees, print QR cards, see who is present right now at each plant, approve corrections, download attendance reports, and see the payable salary of each employee for the month.

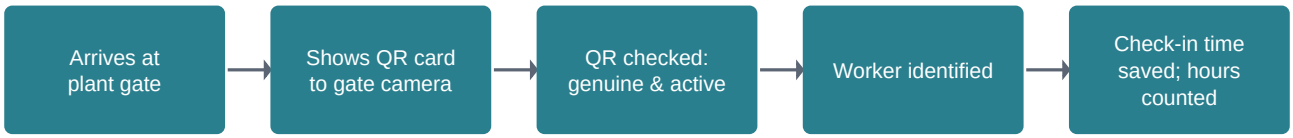
4.1 How it works: permanent employee

One-time: face registered by HR (with the employee's written consent)



4.2 How it works: temporary worker

One-time: HR adds worker and prints a QR ID card in under a minute



When the worker leaves, the QR card can be switched off instantly, so it can never be used again.

4.3 How proxy attendance is stopped

Instead of trusting one thing, the system checks several things together. A person would have to defeat all of them at once, which is practically impossible.

Check	What it stops
The gate device is registered to one plant and stays there	Marking attendance from home or another location
Face match + live-person check (blink / head turn)	Using someone's photo or video to mark their attendance
QR codes are digitally signed and can be switched off	Fake, copied or old QR codes
Employee App: inside geo-fence + own registered phone only	Marking from outside the plant or on a friend's phone
The main server re-checks every attendance entry and uses its own clock	Tampered apps, changed phone time, duplicate entries
Unusual activity is flagged (e.g. two plants 50 km apart within 10 minutes)	Shared QR codes or shared accounts

Every accepted or rejected attempt is recorded with the reason, so HR can review suspicious cases.

5. Core features (first version)

- **Plant setup:** add plants, draw each plant's boundary on a map, register gate devices, define shifts.
- **Employee management:** permanent and temporary workers, contractor/agency details, bulk upload from Excel, QR ID card printing.
- **Face registration and face attendance** with live-person check.
- **QR attendance** with instant activation and deactivation of cards.
- **Automatic calculation:** check-in/check-out, late arrival, early exit, overtime, and total hours for hourly workers.
- **Live dashboard:** who is present, absent or late today, at each plant.
- **Reports:** daily attendance, monthly register, hours per temporary worker (Excel and PDF).
- **Payable salary estimate:** monthly salary payable to each employee, calculated from attendance and the employee's fixed salary structure, with an Excel export (see Section 5.1).
- **User roles:** separate access for admin, HR, plant admin, supervisor and employee; every change is logged.

5.1 Payable salary estimate: how it works

For each employee, HR saves his **fixed salary structure** once (for example Basic, Dearness Allowance, House Rent Allowance, Travel Allowance and other allowances). At month end, the system combines this with the employee's attendance and shows the admin **how much salary is payable** to that employee.

Step	How it is calculated
1. Monthly salary	Basic + Dearness Allowance + HRA + Travel Allowance + other allowances (as saved by HR)
2. Per-day salary	Monthly salary divided by the number of salary days (26, 30 or actual days of the month: to be confirmed by you)
3. Payable days	Days present + paid leave + paid holidays / weekly offs (as per your rule)
4. Earned salary	Per-day salary x payable days
5. Overtime pay	Overtime hours x overtime rate (as per your rule)
6. Deductions	Entered by the admin by hand (for example PF, ESIC, TDS, advance). The system does **not** calculate these.
7. Net payable	Earned salary + overtime pay - deductions entered by admin

Example (figures are only for illustration): Ramesh has a monthly salary of Rs. 30,000 (Basic 15,000 + DA 5,000 + HRA 6,000 + Travel 2,000 + other 2,000). With 30 salary days, his per-day salary is Rs. 1,000. He has 18 payable days, so earned salary is Rs. 18,000. Overtime of 4 hours at Rs. 150 adds Rs. 600. The admin enters deductions of Rs. 2,000. The screen shows **net payable: Rs. 16,600**.

Temporary / hourly workers: total hours worked x hourly rate (plus overtime if applicable).

Output: a monthly salary sheet on screen and as an Excel file, showing for every employee: days present, days absent, leave, overtime hours, payable days, earned salary, overtime pay, deductions and net payable.

This is an **estimate to help the admin**. Final salary payment, statutory deductions and filings remain the company's responsibility.

5.2 Not included in this version (future scope)

- Automatic calculation of PF, ESIC, Professional Tax and TDS, and government-format statutory reports.
- Challan generation and management, Form 16, and tax filing.
- Employee document storage and compliance due-date reminders.
- Real-time alerts about changes in government rules.

These can be taken up in a later phase if required.

6. Data privacy and compliance

Face data is sensitive personal data, so we will design privacy in from the start, in line with the Digital Personal Data Protection Act 2023 and its Rules, the IT Act, and labour-law record requirements:

- Employees give clear consent before their face is registered, in a language they understand.
- We store a secure numeric code of the face, not the photo itself, and it is encrypted.

- Face data never goes to any outside company or service.
- Face data is deleted when an employee leaves. An employee can choose QR instead of face.
- Attendance records cannot be secretly edited; every change is logged with who made it and why.
- Reports follow the formats needed for attendance and working-hour registers.

We will confirm the final compliance list with your HR / legal team.

7. Technology and cost

- Only free, open-source technology is used. **No licence fees, subscriptions or paid APIs**, now or later.
- No special attendance machine is needed. Any tablet, Android phone or laptop with a camera can be a gate device.
- Face recognition runs on normal computers; no expensive graphics card required.
- Hosting: the system can run on a computer/server at your premises (keeps employee data inside the company) or on a hosting service of your choice. We will discuss this with you (see Section 8).

8. Decisions and information we need from you

Please answer every question below. We cannot start building until Sheet Shapers has answered them, because the answers decide how the system is designed.

#	Question	Your answer
Q1	How should temporary workers scan QR? **Option A (recommended):** worker shows QR ID card to the gate camera. Works for workers without smartphones. **Option B:** a changing QR is shown on a screen at the gate; the worker scans it with his own phone.	[ ] A [ ] B [ ] Both
Q2	Should permanent employees also be allowed to mark attendance from their own phone (inside the plant boundary), or only at the gate device?	[ ] Phone + gate [ ] Gate only
Q3	Number of plants, their addresses, and number of gates per plant	
Q4	Approximate number of permanent and temporary workers per plant	
Q5	Shift timings, overtime rules and how temporary workers' hours are counted	
Q6	Devices you can provide for gates (tablet / phone / PC with camera), and for testing during development	
Q7	Where should the system be hosted: your own server/computer, or an outside hosting service?	
Q8	Contact person for questions, and sample (dummy) employee data we may use for testing	
Q9	For the salary estimate: (a) which salary components exist (Basic, DA, HRA, Travel, others)? (b) Should per-day salary be monthly salary / 26, / 30 or / actual days? (c) Are paid weekly offs, holidays and paid leave counted as payable days? (d) How is overtime paid (rate or multiple of hourly wage)? (e) What is the hourly rate rule for temporary workers? (f) Is the salary structure fixed per employee or per grade?	
Q10	A sample salary structure (dummy figures are fine) for testing	

9. Project phases

Phase	What is delivered
Phase 1: Foundation	Plant setup, employee management, QR cards, QR attendance at the gate, admin dashboard
Phase 2: Face & anti-proxy	Face registration and face attendance, live-person check, geo-fencing, employee app
Phase 3: Automation, reports & salary estimate	Automatic calculations, live headcount, reports, and the payable salary estimate with Excel export
Phase 4: Testing & handover	Testing at one plant, fixes, user manual and training for your staff
Future	Full payroll (statutory deductions, PF/ESIC/TDS reports, challans, Form 16), document storage, compliance reminders

Exact dates will follow once we receive your answers to Section 8 and our academic calendar is fixed.

10. Your additions, changes and sign-off

Please write any requirement we missed, or any change you want:

Anything to add or change?

	Sheet Shapers	Project team
Name		Tushar Gawande
Designation		Project Lead
Signature		
Date		